Taking the Bizarre Seriously: Dreams as a Material for Interaction Design

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ABSTRACT

Sleep technologies promise optimised sleep by tracking sleep cycles or by affecting sleep environment. This paper expands sleep technology to address more complex notions of sleep, proposing dream technology as a path for reflective practices that focus on nightly dreams rather than hours slept. Because dreams are bizarre and hard to grasp, we turn to speculative design practices in an RtD framework to investigate the nature of dreaming. This process produced the 'dream sticker machine', a design proposal that visualises a user's dreams with an AI image generator. This tangible dream trace becomes sharable in both public and personal spaces. This proposal formed the basis of a probe to how people might imagine their dreams taking form in the real world, and led to ideas for how the materials of dreams might be taken up as part of design research practice.

CCS CONCEPTS

• **Human-centred computing**; • **Interaction design**; • **Interaction design process and methods**;

KEYWORDS

Dreams, probes, design fiction, interaction design, research through design

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1 INTRODUCTION

We spend about a third of our lives asleep—nearly 25 years over the course of an average lifetime—and much of that time is spent dreaming [21]. Dreams are vivid incursions of the irrational into our lives, and for such a large part of our experience, they are also almost entirely ignored in waking life. Two main concerns drive this study of dreams. First is part of a thread of research in HCI and design that seeks to reimagine designing for the body to move away from ideas of optimization and efficiency to focus on other goals, such as self-knowledge, introspection and reflection, and remembrance. The second concern is with how data and other forms of design material are becoming increasingly complex, confounded, and irrational. We feel that dreams might be an interesting proxy to imagine designing for other kinds of similarly confounded and confusing information. What can design make of the bizarre and irrational?

Designing with dreams takes sleep and dreaming not as a missing part of the life, as gaps or lacunae, but rather as something serious and meaningful in itself. Too often, sleep is something that is instrumentalized or operationalized as something that is needed for the waking life the next day. Our interest is to take sleep seriously, understanding dreaming as something that emerges from sleep and can drive new understanding of life—design rooted in the irrational that can help inspire new ways of thinking about the everyday. Ultimately, this project is about paying attention to and valuing our nightly dreams in new ways through technology. This paper explores potentials in the evolving field of dream technology by asking three questions: a) What current technologies are available for the context of our sleeping time, and what challenges are we facing when designing for the user at sleep? b) What do dreams mean to us and what might be the potential of paying more attention to them in our everyday lives? c) What new dream practices emerge when we engage with and design for dreaming?

Over the course of a four-month design process, we performed autobiographical studies, surveyed sleepers, and interviewed experts to inform a design space that takes dreaming seriously as a material for design, rather than a side effect of 'productive' sleep. It led to a design fiction proposal that uses AI image generation to produce stickers of sleepers' dreams that they can place in the world to annotate urban space with their subconscious feelings and desires. Finally, we evaluated this proposal using a design probe asking how the stickers might be understood and used. These three paths demonstrate how dreams can be used as a design material in HCI and exemplify a designerly approach to a complex design space that is characterised by ambiguity and elusiveness.

2 BACKGROUND

As computational technology has moved into highly intimate spaces [39] , sleep technologies are on the rise as part of the burgeoning self-tracking movement [3]. These hope to optimise sleep by keeping track of sleep cycles or by affecting sleep environment, using various technologies to promise faster falling asleep [27], deeper sleep [69], and offer more data about sleep quality [71, 81]. Although digital devices are already integrated into most bedtime routines [63], how sleep might be handled in HCI is open for debate [4]. This might be because of an ingrained understanding of the potential user being passive while sleeping and therefore not



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initiating direct interactions [55]. Reviews of contemporary sleep technologies [58, 63] reveal that, like many forms of self-tracking, current sleep tracking systems seek to *optimise* sleep, and are anchored in the cultural values of sleeping and productivity as being intertwined. Indeed, this goal has been a large part of how sleeping has been taken up in HCI [54, 55].

This is not fully surprising, as a major motivation for using self-tracking technologies is most often to achieve a goal, and 25% of participants in a study of quantified-self enthusiasts report tracking their sleep [19]. Tracking in the context of sleep could be met by multiple concerns, including having to track sleep continuously with little or no disruption to collect trustworthy data [63]. Further, the assumption around sleep tracking is that sleeping is good, and regular sleep is better. If we cast off the idea that self-tracking needs to be somehow directly productive, and instead take it as a kind of wayfinding (after [84]), we can begin to imagine sleep and dreaming as a data source for design. How might we use self-tracking differently when dealing with incomprehensible things such as sleep and dreams?

2.1 Emerging perspectives on self-tracking

Elsden et al have argued that design trends for quantified self-systems are too focused on solutionism, overly rational analytical insights, and individual goals [30]. Rather than data as for gaining actionable self-knowledge, they suggest we use data to have a better experience of and in the world. To that end, a rich vein of contemporary design research works with self-tracking technologies to understand the body and bodily practices in new ways, ones that reject abstraction of knowledge about the self and instead emphasise interpretation and direct understanding [11, 42, 50]. These cover period tracking and menstrual health technologies [43, 44], gut health and the gut-brain axis [11, 49], among others. Instead of focusing on accuracy and numerical representation of ideas like "healthy" or "unhealthy," technology like these produce data about the body for a novel form of reflection, rooted in personal phenomenological experience [45], and imagine biodata as a material for engaging with ourselves more directly. In this mode, self-tracking becomes a kind of '*self-cultivation*' [11] where we can "(...)explore engagement with and through biodata" [79], potentially taking away some of the performance anxiety or 'tracking fatigue' we see as down-sides by contemporary self-tracking culture [19]. This shift towards more qualitative, domestic, and reflective self-tracking practices in turn inspired us to design with dreams themselves, rather than simply tracking sleep.

2.2 Dreams as a space for design

Dreams offer a rich starting point for engaging with the irrational from a kind of altered state [61]. Each night as we sleep, we experience things that are entirely imaginary, tell ourselves stories that are related to our lives in ways that are both tenuous as well as deeply situated. Experiences from dreams can affect us in our waking lives, and they have been a preoccupation of many different cultures, philosophers, and psychologists—from the Dreamtime creation stories of indigenous Australians [83], to dreams as oracles to the Greeks and Romans [7], and as fodder for psychoanalysis for Freud and Jung [31]. There are many theories as to why we dream: as a way of consolidating memory [28, 72], as a mechanism for self-organization [53, 88, 89], to generate and spread cultural diversity [64], as a by-product of the brain's physiology [23, 41], as a means of simulating possible future threats [75], among others, but no explanation has proven conclusive. Dreams are hard to describe, hard to remember, and hard to define—both for researchers and the sleeper—and can be interpreted in various ways. This makes the dreams and their contents hard to work with as a material for design, as they will always be told based on subjective recall and never as objective, external observations. That said, dreams historically and culturally contain a great amount of value and significance in the fields of art, psychology, and medical science [29, 61, 64]. However not all dreams can or should become material for design. Dreams are uncontrollable, meaning that they can transgress cultural norms in ways that are broad or narrow, making them unpleasant to reflect upon. They can include sexual content that might be inappropriate or reflect traumas.

Partially based on this uncontrollability, commercial dream technology is relatively narrowly focused on helping sleepers achieve lucid dreaming [16, 74], or dreams where the dreamer can take command of their dreams [62], and apps that offer interpretations of dreams [36]. As with other forms of sleep technology, these too are implicitly goal-oriented and aimed at optimising sleep—including achieving lucid dreaming—rather than exploring how dreaming itself might become a part of people's reflective everyday practices. To some extent, this gap has been taken up by academic research. *Dormio* is a project of MIT's Media Lab that uses the hypnagogic period between wakefulness and sleep as a resource to extend creativity [18, 38]. Brand et al have used an AI-based dream interpretation technology called *Dream Streams* as part of a suite of speculative products to imagine how AI might become part of broader introspective practice [12, 13]. *The Dream Painter* is a project that uses a robotic arm to draw pictures based on dreams that are described to it [17]. Other entry points to this space come from speculative and critical design [6]. *Luce*, for example, proposes a wearable pendant that captures dreams as part of a 'quantified dreaming' project [66]. Together, these articulate another agenda for dreaming that moves beyond the control, calming, and tracking agendas that are hallmarks of sleep optimization, one that remediates or represents dreams for knowing yourself differently. Building on these, we see potential in using dreams in other ways such as for sharing, connecting with others, for simple entertainment, or working with the experience and content of one's dreams in new ways.

3 METHODS

We used a research through design (RtD) approach [34, 90] to develop this project in an open-ended fashion. RtD uses a mixture of research and design methods in its process [34, 59], together allowing designers to reflect on issues and understand the broader possibilities of an issue or agenda [26]. This project took place in Copenhagen, Denmark, and all authors are of Northern European descent. Dreams and the role of dreaming in society have different cultural connotations in other locations, and this project does not intend to offer generalizable findings about dreaming as such.

Table 1: Autoethnographic findings.

Task	Findings	Pros	Cons
Write down dreams on the phone after waking	Quickly became a chore; frustrating; dreams slip away while typing; blue light irritating at night	Accessible, private, easy	Bright screen, slow, requires the phone, limited
Audio record dreams after waking	Awkward at first, but became routine; uncomfortably public to externalize dreams; unsure of value from listening to them later	Quick, easy	Exposing, awkward, can disturb others
Tell someone about a dream every day	Hard to remember dreams later in the day; connection sparked with others–people surprisingly interested in talking about their dreams	Personal, cultivates a space for talking about dreams, can help dreams become better remembered	Hard to remember dreams often, forced, too personal, cannot be revisited
Draw dreams with pen and paper	Quite demanding and difficult; forced a prioritisation of dream elements, helped us to notice new details about the dreams	Remember a dream better, new perspectives, fun	Impractical, time-consuming, pressure to do it well, dreams hard to remember in enough detail

Rather, it seeks to investigate dreaming within our own cultural milieu to develop new ways that dreams might be taken as a material for interaction design.

To engage with dreams as design material or as part of a design space, we needed to first understand current dream practices—what perceptions people have and uses people near us make of their dreams. To that end, we employed a variety of approaches to learn about dreams and dreaming as a possible design material. We used autoethnographic methods [25, 68] to learn about our own sleep practices and experiences with remembering, recording, and sharing dreams to provide a foundation to start speculating towards different kinds of dream technologies [6]. To obtain more general information, we deployed a questionnaire [8] to 100 people between 15 and 60 years old recruited via social media and in person through snowball sampling [52]. This survey included a mix of open-ended and multiple-choice questions asking about sleep quality, habits, technology use, and their relations to and thoughts about dreams and dreaming. For more pointed inquiry into sleep and dreaming, we interviewed three experts: a dream therapist, a dream practitioner, and a neurologist. These were semi-structured interviews [76] using common interview guides with added questions for each expert that related more specifically to their field.

4 KNOWING DREAMS AS A MATERIAL FOR DESIGN

To orient our design research process, the first and second authors (26F, 28F, master's students in interaction design) began with four weeks of autobiographical methods [25, 68]. These methods were meant to provide direct experiences of remembering and reflecting on our dreams, including how to record and keep them as a kind of data material. First, we experimented with simple ways of recording dreams. In the first week, we wrote down the dreams on our phones upon waking, and in week two, we would tell the dream story into the phone's voice memo app. Inspired by [47], which described how one could develop closeness with others through disclosing dreams, in week three we would find someone to tell our dreams to each day. Finally, in the fourth week we would draw our dreams as a means of hopefully understanding and interpreting them in new ways [60, 78]. During these four weeks, we took notes on our experiences and used thematic analysis to cluster how we understood each of the approaches [14].

While there were advantages and disadvantages to each of the tasks (Table 1), they led us to interact with our dreams more frequently and richly and helped us learn how to record and remember dreams, as well as why one might want to do so. Notably, we found that talking our dreams into a recorder was a particularly convenient way to record them, and that sharing our dreams with others was a good way to make a new kind of connection with others that we didn't expect initially. Visually representing the dreams with paper and pen to make them last longer helped us to get new perspectives on our dreams. We found potential from these activities in designing for dreaming that focuses on remembrance and introspection, as well as making dreams more material to support those practices.

4.1 Survey

While the autoethnographic activities indicated some value in recording dreams, we also wanted a broader perspective on what people thought about their dreams, and how those ideas of dreaming intersected with how they understand their sleep. We used a qualitative questionnaire asking participants to share their reflections, experiences, and opinions surrounding dreams, sleep, and technology. To recruit participants, we shared our questionnaire on a university forum, on our personal social media profiles, as well as with friends and family, asking them to share it in turn their networks as a form of snowball sampling [52]. In total, we got 100 responses. Most of the respondents were between the ages of 21-30, but ages from 15 to 60 years old are represented. We analysed the responses thematically to identify common threads across the respondents [14]. While this sample is certainly not statistically representative, we hoped to get a wider perspective on sleep and dreams that might lead to insights generative for designing.

4.1.1 Sleep matters. 98% of our respondents sleep between 6-9 hours daily. 73% of people sleep exclusively at night whereas 23% take naps as well as sleeping at night. This underlines that we are not getting to know extreme sleepers, but standard sleepers [55]. 42% of the people we asked thought dreams were mostly something fun, while 40% associated dreams with something scary and unsettling. Respondents were divided between seeing their dreams as something personal reserved for themselves (26%), or as an exciting topic for conversation (24%). 21% of people thought they dreamt more when they slept well and 14% of people thought they dreamt more when they slept worse, with 43% feeling that sleep quality did not affect their dream frequency.

Our respondents described many different types of dreams. *Good dreams* were generally seen as having a positive impact on a person's mood, while *nightmares* affected most people negatively. These could bring judgments into real life if a friend were mean to them in a dream, and some respondents felt that their mood was affected the following day. Another type was a *productive dream*, where a dream could help them solve problems they couldn't while awake: *"last night I dreamed how to handle a current task I have [at work],"* or "*I made a drastic decision that changed my life for the better based on a dream I had.*"

41% of the participants interested in understanding and interpreting their dreams, but that it was not something they thought about often. 28% of the participants felt it made sense in their lives to reflect upon their dreams: "*More attention to being stressed, for example that I need to take care and think things through,*" or "*I have used them to reflect on my own experiences and opinions on things that have happened.*" Others mentioned how they used their dreams to reach a specific goal or decide on an important choice in their lives: "*I made a drastic change in my life that changed it for the better because of a specific dream;*" "*I had panic dreams related to my job which was a symptom of stress, so I decided to study part-time for a while*"; "*I've used them to reach a specific situation/goal that I had dreamt of.*" For some people, dreams could be used to indicate the state of their mental health or overall well-being: "*I may use it as a reflection of how I feel;*" "*I use it to figure out if I'm too stressed in my everyday life, and then I try to clean up what stresses me out;*" "*I have the same dreams over and over if I'm too stressed. I take it as a warning sign that I need to relax more.*" Some people said that they used their dreams to get to know themselves better, or to reflect on their sexuality, relations to other people, and so on.

4.1.2 Remembering dreams. We also asked them about techniques for remembering dreams. Some people wrote their dreams down or recorded them on their phones, and some wrote them down on paper. Some people shared their dreams with friends, family, or loved ones to better remember them. One respondent said that they held onto the feeling they had when they woke up to remember the dream, and several people said that their main technique was to 'think them through'. Lastly, a person also mentioned using manifestation before sleep as a tool. We found that people also had very different reasons to document their dreams. Most people simply did it for fun or to drive conversations, for example bringing them to their therapist to discuss and unpack. A few people wanted to train their memory to better remember future dreams. Some wanted to remember their dreams so they could revisit them later.

We asked people about 'visualizing dreams', and more specifically, how they might want their dreams visualized. Most people suggested some form of visualization like watching their dreams like a movie on a screen. Two people said that emotions and individual experience were extremely important for the dream: "*I wish people could see them like I remember them in my head. Even if I could visually show my dreams, many of them have a specific meaning because there is an emotional element specifically related to me. Therefore, if I had to show my dreams, I would like people to see it from my point of view, including the emotions.*"

4.1.3 Sharing dreams. 54% of respondents told their partner about their dreams, 52% told their friends and family, and 31% told their dreams to the person they dreamt about. Nine people would tell them to anyone willing to listen, and four people said that they told people about their dreams, but people did not care. One person had a bad experience of sharing dreams and did not share their dreams anymore. Five felt they were too private to share.

We asked people whether they would show their dreams to other people if it were possible. 34% said they would, 6% said yes if it was anonymous, and 42% said they were unsure. When asked whether they were interested in hearing about other people's dreams, 57% said that it depends on whether they know the person well, and 41% said they were interested in hearing about other people's dreams—but 16% responded they are only pretending to be.

4.1.4 Sleep technology. 55% of our respondents had not used any kind of sleep technology, while 34% had and 11% were unsure. In general, the people who had not answered that it was because they had never considered it, and some were worried that it might affect their sleep in a bad way or that it would be more disturbing than helpful. For the people who had used sleep technology, 41% did so occasionally, while 35% used it every day. Most used it to either fall asleep faster, or to get an overview of their sleep habits. Some people saw *sleep tech as disturbing/invasive* and would rather not bring technology into the sleep situation in fear that it would influence their sleep quality in a bad way. Most people only knew or tried sleep technology in the form of either apps or smartwatches and had some concerns about the data being hard to decipher or employ in useful ways, and some also had concerns about how devices collected bodily data. Some people said they would never use sleep technology as they saw it as being *only for solving problems.* A few people also mentioned that sleep technology was not for them as it was simply *too expensive.*

We made a list of initial ideas for what purposes a dream technology could have and asked which people could imagine using. Most people (58%) wanted to use technology that had the goal of presenting data about the themes that occurred in their dreams. 48% also said that they would be interested in a technology that made it possible to revisit dreams. 47% were interested in the correlations between how many dreams they remembered and how many hours they had slept. Another 47% would like a technology that could help lucid dream. 33% of the participants were also interested in a technology that could visualise or manipulate dreams.

Overall, the survey results led us to believe that there was more interest in technology for dreaming than we had initially expected. Most intriguingly for us, 76% of respondents were either interested in or possibly interested in sharing their dreams somehow, and

most of our respondents were interested in a system that could give them thematic data about their dreams, with more moderate interest in visualizing or manipulating dreams.

4.2 Interviewing experts

We wanted to augment these broad perspectives with ones from people who have a special relationship to dreams and dreaming. Dreams are used in many different fields and can be viewed through a variety of lenses and from many different viewpoints. Seeking a wide range of perspectives, we were especially interested in how dreams might be included in psychological treatment and care from an interpretive perspective, and a more scientific and medical perspective on dreams and dreaming. The first led us to an art therapist who uses dreams as part of a therapeutic practice, and the second led us to a professor of neuroscience at a local university who focuses on the biological basis of dreaming. Finally, a Google search for "dream expert" led us to an author and who is interested in how dreams can be used and applied in personal as well as professional contexts. This offered a new on dreams to augment the more biological and psychological areas with a more social and practical area.

4.2.1 The dream therapist. Karen is an art therapist whose methods concentrate on dreams as offering insight into the subconscious in both children and adults. Her research indicates that forgotten memories can be revealed in dreams and help drive therapy. Karen considers dreams as the product of experiences, emotions, and traumas. Working with them can help introspection and understanding our reactions or feelings, and given the right tools and training, can be a catalyst for problem-solving.

To train this ability, Karen believes we should concentrate on detecting what dreams offer. Writing your dreams down, talking with others (such as a therapist), or drawing them all make it easier to access their contents. When drawing dreams, Karen says, you get a more intuitive interpretation of your dream through its visual aspects: *"(..)when you sit down and draw a dream, some items and persons will inevitably become larger or smaller (..) So, one's intuition is used when working with and drawing the dream. Then you can examine the drawing and from that support what has happened in the dream."* When using dreams in an introspective practice, she believes that intuition is the central part—what sparks your interest most in a particular dream.

Another point she emphasized is the connection between the body and the mind, particularly with respect to dreaming. To recall a dream, she works with the body as a sensing organ. Using meditation to get back to the feeling you've had in the dream, coupled with the dream's drawn representation, can be a tool for finding important elements in it. Seemingly random interactions can be a sign of your subconsciousness at work. When asked if she could imagine any technologies for dreaming, she worries that this bodily aspect of dreaming might be neglected or reduced, especially given current device categories like smartphones and smartwatches that bring bodily knowledge outside of the self.

One possible design angle from this project is that dreams can be seen as a mirror of both the subconscious as well as the body's reactions in everyday life. Paying more attention to bodily senses or emotions might be productive when working with dreams. Our senses can make us recall dreams in the middle of the day because of a scent, a person, or a place, for example. A constraint is that dreams are wholly subjective and individual. Replicating or translating their importance through technology may be impossible, even if it were somehow desirable. This means that AI dream interpretation, for example, might not be useful—at least in a therapeutic sense.

4.2.2 The neurologist. Birgitte leads a neuroscience research group at a local university focusing on exploring sleep biology and how sleep and dreams relate to one another. She described how we can separate our sleep into four phases: N1, N2, N3, and REM sleep. REM seems to have a strong relationship with biological health: while waking up several times a night does not count as bad sleep, not getting in the REM state affects our frontal cortex and brain in general. The REM state is also important to enter for more vivid and complex dreams, and you particularly want to wake up in this phase if you want to remember dreams. Birgitte describes our brain as not programmed to remember dreams: "*While dreaming at night, [dreams] are not getting stored in the normal memory. If you want to remember a dream, you must transfer it to the waking memory processor as soon as you wake up. Thus, you only remember the dreams you replay while awake.*" To do this, Birgitte recommends that you wake up slowly and either speak to someone about your dream or write them down.

The most substantial scientific explanation for dreaming, in Birgitte's opinion, is that we dream to prepare for possible future scenarios, or in her words "*(...) for building better models of the world.*" Part of this lack of certitude is that dreams are extremely hard to measure. To analyse a dream biologically, it is necessary to measure the brain's electrical activity, and this might seem invasive to an everyday user if it had to be incorporated into a dream device. Lucid dreaming is likewise difficult to measure ethically, as it is unpredictable, rare, and most common among children.

This interview led to three main insights for using dreams as material in HCI design. First, measuring dreams in real time is uncomfortable and possibly invasive, meaning that for this project dreams will mainly be taken as a design resource after the fact. Second, that the brain is not wired to remember dreams, and it takes concrete effort to store the dream in longer-term memory. Finally, both moving away from real-time dream design as well as the difficulty of assessing lucid dreaming underscores that lucidity while dreaming is not a sensible target for this initial design inquiry.

4.2.3 The dream expert. Michael is an award-winning speaker and best-selling author on dreams and dreaming. He has worked 11 years as an expert on dreams, writing books and articles and giving presentations about the potential of working with one's dreams. Echoing the findings in our questionnaire, Michael tells how dreams are mood-regulating, and for some might affect their entire day or more. Further, he describes that dreams have possibilities for problem-solving in both personal and professional life. Talking or thinking about your dreams might bring you to acknowledge something deeply personal about yourself that you might otherwise miss. To him, dreams can produce concrete images from our subconscious where intuition, creativity, and problem-solving are conjoined—it's "*(..)a bit more magical if you'd like*", as Michael says. He discounts symbolism as a strategy and tool for translating a dream, rather than giving meaningful answers. Interpreting the



Figure 1: An overview of landscapes of imagined future dream technology.

dream, articulating 'how *you* see it' and what it evokes in the individual case is what is important, rather than believing that we can understand a dream without getting the dreamer's deeper conception or perspective. Simply remembering your dreams is not enough: "*Different methods for remembering dreams are one thing, but the major breakthrough comes when you start working with them*," for example in dream-based group therapy. Michael believes the combination of dream reading's bad reputation as a pseudoscience as well as our general lack of time makes it hard to imagine dream groups becoming popular again—it's too time-consuming, and too hard for people to focus.

For this project, this means that there are no generalisable methods to interpret a dream, and the same act or symbol in a dream can have different meanings for different individuals, contributing to dreaming's magical quality. The perception of dreams as private vs. shared is interesting to examine further. Where does a dream go from being private to becoming a collective matter? The survey indicated division as to whether people are interested in listening to or telling others about their dreams. Dreams are private, subjective experiences that mirror our inner world. There is potential in opening up about your dreams, but it takes time to invest in other people's and trust to share your own.

4.3 Transitioning to design

The above section describes several ways we gathered information on how dreams are viewed, their potential in a design space, and how dream practices are managed in people's everyday lives, both from different dream experts and autobiographical methods. One surprising finding from this work is that dreams are something that people are interested in talking about, or at least could be. *Sharing dreams* is something that most of our questionnaire respondents are interested in, and the idea of group dream work mentioned by Michael, the dream practitioner, draws out a social role for dreams to play. Another strong theme is the idea of *putting dreams to work* for you somehow, whether as an avenue for problem-solving or a way to know yourself better. Much of the autobiographical component of this research was around *recording and materializing dreams*, and how best to record them. Karen, the art therapist, describes representing dreams as an intuition-led way of paying attention to what seems most important visually, and how those visuals come to represent aspects of the self. While different people had different reasons to record their dreams, part of what makes them so provocative is in the difficulty that holding onto them presents. Birgitte, the neurologist, says that it takes special effort to move dreams into the longer-term memory. Here, we may ask what interaction design and HCI can do to capture dreams. Finally, there is a theme that dreams *have biological origins and relations* to the self in ways that are not fully understood. linkage is somehow both real and not quite true: dreams clearly come from the sleeping brain, but also offer access to a wealth of new and generative concepts that have a bizarre, almost mystical quality to them.

5 DREAMS AS A DESIGN MATERIAL

Inspired by the themes above, namely *sharing dreams*, *putting dreams to work*, *recording and materializing* dreams, and the *biological origins* of dreaming, we generated a variety of design concepts for recording dream content [59, 85] (Figure 1). The designs each explored different ways of thinking of, materialising, and 'translating' dreams. The imagined designs were a way to explore different dream practices and encapsulate dream diversity. We specifically avoided becoming apps to give dreaming a material presence [46, 48] in the lives of dreamers. The designs explored the notion of using dreams as material for interaction design through different approaches, such as embodied interaction; as something private; something to be shared; something to be manipulated; something to be interpreted; and ultimately as something to prompt reflection [40, 77].

5.1 Design concepts for dreaming

One direction for these concepts was through *visuals*. The '*Dream Orb,*' for example articulates a mood or mental state based on the contents of the dream. This concept is not meant to be seen as something scientific but more like guidance or entertainment way to see what your dreams could represent. Like a mood ring that we knew would not be truly accurate—in fact it would be wrong most of the time—it still made us reflect on our mood as it changed. Through '*Dream Code*' we imagined a more abstract way to represent dreams as simple colours which might seem random at first but may contain an identifiable pattern if used over time. Beyond the simple colours, we also worked with dreams as something more representational. '*StickIt*' explores visualising dreams through AI image generation to

make stickers that can be placed wherever a dreamer wants. '*Dream Bud,*' a dream-based Tamagotchi, changes and grows depending on the dreams you tell it.

While visuals may be a common way to represent dreams, it also made sense to consider dreams as fundamentally having or being *stories*. '*REMMY*' is a toothbrush you tell what you remember of your dream while brushing your teeth in the morning and receive a possible ending. '*Teddy Talk*' translates your worries and fears into nice bedtime stories so a troubled sleeper may have better dreams. *REMMY* and *Teddy Talk* explore dreams as something private and personal. As a contrast, we also worked with dreams as *shared stories*. Instead of thinking of dreams as something private, these designs instead explored dreams as something to be shared with others, in public or among friends. '*Dream Reflection*', for example, is a mirror that takes spoken dreams from a household and prints you a fragment of another dream it was told. More abstractly, '*Maren*' is a statue in public that receives typed dreams via Bluetooth. It moves when it receives a dream and displays stats on topics and themes on a small screen. The '*Public Dream Keeper*' is a series of locations where you can tell it a dream and listen to the dream of another. Finally, '*StickIt*', described above, bring dreams from your home into the world via AI-generated stickers.

The exploration of dreams in relation to *bodily sensation* mostly centred on the sleeper through embodied interactions around the effects of dreaming. '*Warm-hearted*' is a necklace that rests on the chest and warms during both REM sleep as well as at random points throughout the day; '*Pilowww*', which translates your dreams into soothing vibrations; and the '*Black Out Mask*' identifies and tracks which centres of your brain had had activity during sleep.

All these explorations worked with introspection as a goal in some form, in line with the turn in self-tracking above. Some of the designs worked within the notion of collectiveness, where a user might think about how their dreams affect their lives by connecting and sharing dreams with others. Some of the designs explored dream interpretation, encouraging the user to create the meaning of their dream for themselves. While we did not want lucid dreaming to be a primary focus of this project, as it is what the landscape of contemporary dream technology is already focused on, we also did not want to completely neglect it, either, as it is a frequent goal for sleepers in our survey. To that end, we worked with the aspect of lucid dreaming through '*Warm-hearted*' and '*Lucely*' which both have elements that function as *'reality checks'*—interactive qualities that might cause a dreamer to enter a lucid state [1].

5.2 The Dream Sticker Machine

Reflecting on the variety of proposals above, we ended up wanting to work more deeply with the sticker machine. It combined many of our core findings from the background research, including representing dreams to better remember them, sharing dreams with others to form new relationships with others and yourself, and offering opportunities to explore new forms of technology. Moreover, we figured that stickers could be a provocative way to engage with dreaming. This concept was inspired by street art practices that wish to reclaim art as part of a social practice that exists outside the gallery space, particularly in Copenhagen [5]. The sticker as an object and genre has a range of uses and purposes, from communicating activist standpoints, sharing artistic expression, or being part of marketing campaigns [73]. In Copenhagen, placing stickers in public spaces and on public infrastructure is not strictly legal. However, there is a vibrant communication medium present: traveling football fans stickering over local the club's, political expression and reactions from those that disagree, and promotional material and advertising all vie for space on light poles, utility boxes, and walls creating a "a specific layer of urban communication, especially in areas of pedestrian transit within a city" (ibid). As a design concept, The Dream Sticker Machine invites a user to reflect in the relationship between a dream and its image, how they may be used to connect with others in the world, and positioning your



Figure 2: Sketches for a Dream Sticker Machine prototype and how dreams might be shared in public.



DREAM WITH FANTASY ELEMENTS
I was in an underwater world because I had used the move 'dive' from Pokemon. I was there with my boyfriend but suddenly I lost track of him as he went to the toilet and never came back.

A REALISTIC DREAM
I was working at my job sitting in the reception on my computer. The day felt very long and all the tasks I had were difficult, as if I had to start over learning how to solve them. There was a deadline that I was stressing about at the same time, and the tasks just piled up.

A NIGHTMARE
I was trapped in a giant labyrinth with my best friend. We were running around feeling scared as there was a monster chasing us. We suddenly reached a dead end and the monster was closing in on us.

Figure 3: Investigating AI image styles. We tried to find a suitable style for portraying a dream by using ourselves as test subjects. We tried out six different styles for each dream we could recall and chose the best-suited style (in purple).

dreams spatially through sharing them—and possibly remember them better over time. Conceptually, the sticker machine seemed rich: It did not intend to solve a particular problem for sleepers, but rather created an opportunity to understand and extend dream practices in new ways.

The dream sticker machine is a design fiction [9] (Figure 2) that imagines a system placed near a sleeper's bed. Before a dream has faded away after waking, a user using a conversational interface to tell the device a dream, generating an image of the dream from the story it was told. A user says a prompt like 'I had a dream…' to ready the device to listen, and after hearing the dream, the sticker is printed. Inspired by smart home technologies like Amazon Echo or Google Home, the product language and situation provide a perceptual bridge [6] between the sleeper and the speculative situation of dream recording and representation. The imagined uses of the Dream Sticker Machine are detailed in the descriptions below. These became touchpoints during the design process and gave us ways to articulate the role of the system, building and influencing how we imagined it existing in the world [22].

Let your dreams be visualized. It's hard to give a detailed account of a dream. Images you recall while waking quickly disappear in the day, and situations in your dreams seem absurd for others to imagine. This device gives you the opportunity to print a picture of your dream—or at least how an AI interprets it.

Share and start a conversation. Did you dream of someone you know last night, and want to tell them in a discreet way? Hand them a sticker of a dream and spark a conversation! Dreams might bring your friendship to new hights. Maybe your friend has their own opinion or views for what the dream is about?

Create dream territories. Decorate your neighbourhood by placing your dream stickers on trashcans, lamp posts, or bus stands. Place your nightmare somewhere you will never visit again or take a walk and see other people's dreams. Placing the stickers in your community reclaims public space, declaring that dreams are worth discussing.

5.2.1 AI-generated dream imagery. We used generative AI to represent dreams as images. AI is a heavily-debated and quickly-evolving way of processing and representing information [65], even as questions swirl around the labour rights of artists and how datasets trained on corpuses of copyrighted materials can be ethically used to make new products and services [51, 57]. For this project, we saw an interesting coupling between AI and our dreams, both metaphorically and practically. Though dreams have been tied to reflecting on the past, some people also see it as a potential way of predicting or envisioning the future. Both generative AI and dreams involve manipulation and cross-referencing historical data as part of complex systems. Just as dreams present 'bizarre' imagery [20] rooted in our own experiences, AIs create images in unexpected forms and expressions based on large image databases [91], and indeed early image production models were referred to as a kind of 'dreaming' [24]. AI-generated images and dreams share a sense of human or real-world expression at first glance, but there are often gaps or things that do not make sense at a closer look. While AI image generators have been accused of being notably bad at drawing hands, for example [80], scenes, people, or artefacts also take weird or unexpected forms in our dreams. Both forms of image rendering are not always what they seem, but ultimately stem or derive from something that is—or was—real.

We chose to use Hotpot AI [2], an open-source and free image generator that offers a number of different styling options. Ideally, we wanted something that could encapsulate the overall feel and story of a given dream. We experimented with different styles ranging from realistic to fantasy and anime influenced. We decided to use three different types of dreams we recalled as prompts to explore the different styles and feel of the possible AI images. The results are seen below (Figure 3).

We tested portrait, landscape, and scenery image renderings to see how different emphases of the dream content played a role in the output. The choices we made were strongly influence by the different framings and categorizations that the tool offered us. Ultimately, we chose a quasi-realistic and cinematic style to represent the dreams. The images the model produced were often quite different than the dreams as remembered by us describing them. Obviously, the image created by the model could not replicate of the exact place the dream took place or illustrate a real person as they

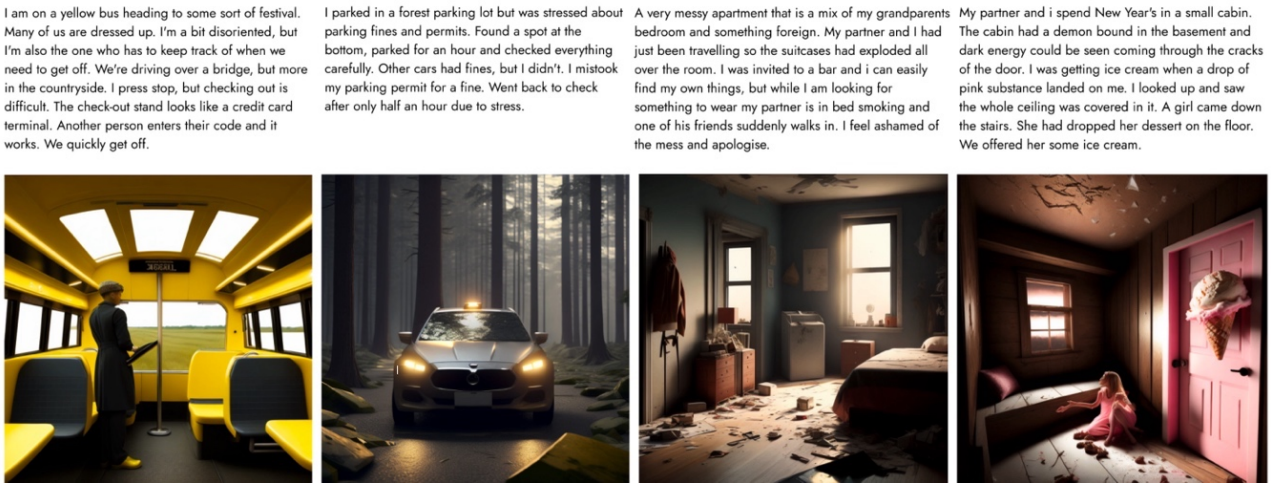


Figure 4: Generated dream images and the descriptions that created them.

were in the dream. This led to a distant feeling of the AI somehow 're-dreaming' for you. The heavily stylized cinematic rendering of the dreams contributed to this as well. Dreams became mediated into a media form that might make more sense to consume than the more prosaic imagery that might be truer to actual experience. Sometimes, the AI disregarded part of the prompt and told its own story—hallucinating while representing dreams [82]. We also found that there was an important balance between fantasy and realism. We didn't want a dream to feel too far from the real world, but at the same time, we wanted to maintain the magic of dreams in their depiction (Figure 4).

5.2.2 Stickers. We chose stickers as output because of their flexibility. They are desktop stationery, toys for small children, and collectibles, and from these perspectives offer a fundamentally playful way of interacting with dream contents [35]. Beyond this, though, stickers have taken on a number of roles in the urban sphere, from communicating activist standpoints, sharing art, and being part of marketing campaigns [73]. Ultimately, stickers are an inexpensive and simple way for people to set a mark in common space. In this concept, stickers suggest a way for people to start conversations about dreams. We hope people will want to bring aspects of their private lives into the public sphere to change the way we communicate about dreams, shifting dreams from being intimate or revealing to becoming empowering or inspiring. Idealistically, the stickers themselves may help propose a way for people to express deeply private aspects of themselves in public in a playful way.

6 EVALUATION

The second phase of this project, above, developed a more concrete design proposal, the *Dream Sticker Machine*, that uses AI image generation to make stickers based on the dreams of its users. To understand what others would make of it, and to see what aspects of the sticker concept seemed promising, we recruited 8 participants (ages 25-54; 6 female, 2 male) interested in dreams to experience a design probe [10, 33, 67]. This took place through an open call on social media. As the design concept was partially imaginary, this probe used wizard of oz methods [15, 37] to simulate a fictional service that creates stickers from dreams for participants to place and make sense of as they might. We chose not to explain the concept around the actual device as a design fiction, as the focus in this probe was to test the premises of the stickers and the feeling around sharing materialised dreams in this format.

6.1 The sticker probe kit

To prepare the probe kits, participants had one week to send us three recent dreams. The dreams were then processed through Hotpot AI. We chose not to set limitations on the length of a dream description as we wanted their telling of the dream to be as accurate and intuitive as possible. Hotpot AI has an input limit of 400 characters, meaning that we needed to shorten some of the submitted dreams, and translated them to English to be interpreted by the system. Preparing the dream imagery inevitably positioned us as editors or mediators of the dreams. While not avoidable in this context, it is an important deviation from the design concept as imagined above. We imagine the system as producing images from dreams without human intervention, meaning that the dreams we received in the probe might be curated by our participants. It stands to reason that the dreams we received were likely somewhat more innocuous than they might be in a fully automated environment. At the same time, it means that the dreams we received were dreams that the participants were comfortable sharing with us. The final probe kit consisted of an envelope containing stickers of their recent dreams, as well as two instruction cards. These had QR-codes on the back leading to two surveys for during different stages of the probe (Figure 5).

The first survey asked them to first look at the stickers and answer a couple of questions on their immediate impressions, such as whether the shape and size seemed appropriate, and if they could identify their dreams in the stickers. We wanted to understand how they felt about seeing the stickers, as well as how they felt having an AI interpret their dream. Afterwards, participants were given a week to place the stickers anywhere they wanted. We chose to use a



Figure 5: The probe kit was delivered to participants in an envelope, including stickers from recent dreams, instructions for participants, and links to fill out initial and reflective surveys.

week for this study because it both made participating in the study seem more manageable than a longer-term commitment, as well as gave some urgency to participants to place the stickers quickly. Once the stickers were placed, we asked participants to take photos of where they put them. The pictures functioned as a link to their experiences [67], but we were also aware that the pictures might need explanations. In the second post-activity questionnaire, we asked them to elaborate on their experience of sharing dreams in this way, where they placed their dream stickers, and why they chose those locations. The relatively short study does mean that the findings from it are limited: rather than changing practices around sleep or dreaming in the longer term, it demonstrates the acceptability and possible promise of this mode of engaging with dreaming for design. Finally, as noted above, while there is a vibrant communication medium in Copenhagen through stickers, for some participants there may be some hesitation to place the stickers in public places for legal reasons.

6.1.1 Initial responses. Most of our participants liked the stickers, but others mentioned how they had a 'gloomy' or 'dark' ambiance. While many thought it matched their dream experiences to some extent, others found them too specific in their representations. *"[They are] A bit darker and not as abstract and unspecific as I imagined, but dreamy in their vibe"* (P8), *"They look amazing! They encapsulate the feelings of the dream very well"* (P1). Our participants commented on how the sticker might be larger, smaller, or more rounded instead of the 5 centimetre square we had produced: "*Shape is fine, [I] wonder if it were a more deformed shape it [would] match the blurriness of my dreams more"* (P4). For many, the size we offered them were sufficient to represent the dream adequately, and one participant mentioned how it felt more intimate for them: "*The size is small, but that's actually okay. In a way, the small size makes it like I can have the dream in my pocket with me. It becomes intimate in a way it wouldn't have been if it were bigger"* (P7). Only one of our participants did not find it clear what dream was represented by which sticker. For all others, it was easy for people to identify their dreams by the visualizations. When we asked them whether the stickers represented the dream they had had, we got varied responses: "*Well, it's a bit weird because I feel it represents a caricature of the dreams. Like the very visual representation of something that gets the idea there but is not how it looked/felt in the dream"* (P2), "*they represent the feelings and some of the elements in the dream, but they are not exactly what happened"* (P1).

With few exceptions, people felt that the stickers represented their dreams nicely, and that they were reminded of their dreams—or at least elements from them—when presented with them: "*I think my dreams are usually very feeling-based, but looking at the images I can actually remember more of the visual parts, which is something I had a really hard time explaining the first time"* (P7). It differed which feelings people recalled from their dreams, but people were especially able to recall disturbing feelings from the stickers. This might have something to do with nightmares being more memorable than neutral dreams, but these also made us aware of and question whether the AI image style was enhancing 'bad' rather than 'good' feelings in the dreams: "*Because the images are a bit dark, they really enhance the sad/bad feelings. I don't know, I just imagine if it had interpreted a forest as a light green thing, it would have been a happier memory?"* (P7) *"The uneasy feeling, eerie, mysterious... and quirky"* (P8). *"I feel like the train track one is most evocative. I feel the emptiness of that world and the feeling of traveling along the tracks"* (P2).

The last question we asked them was about how they felt about having an AI recreating their dreams. Nearly all respondents thought was interesting, and some noted how it sparked their curiosity about dreaming and made their dreams into stronger memories: "*weird, but also piques my curiosity. I guess because it's a way to re-engage with them, which otherwise I would not" (P2). "It's quite fun to see my dreams in pictures. I kind of feel like it's turning into my memories instead of just dreams" (P3). "I really did not expect it to be such a tool in focusing on the visual part of my dream rather than the feelings" (P7).*

6.1.2 After deployment. After a week to place their stickers (Figure 6), our participants returned with feedback. Like their dreams, people tied different meaning and importance to their dream stickers. People's experiences and rationales for putting the stickers where they had varied. Some people felt they needed to put them someplace special to not waste the stickers: "*The first one I put on my PC immediately as I felt it would fit nicely with the aesthetic on it. And the other two I didn't quite know, so I put them on the fridge and then in the future if I found a better place, I will put them there. I don't want to waste them just anywhere"* (P1). "Others liked the playfulness of placing them as you go: *I love the randomness of just doing something, so I did"* (P4). Still others felt excited to place them somewhere they could revisit them and remember them: "*I felt attached to my dreams, so I mostly wanted to put them on personal stuff so I can remember them"* (P5). Finally, one person mentioned that they did not enjoy putting up the stickers in public as it gave them a feeling of doing something illegal: "*It made me feel uncomfortable, because it is not really allowed. I felt a bit like a graffiti artist"* (P6).

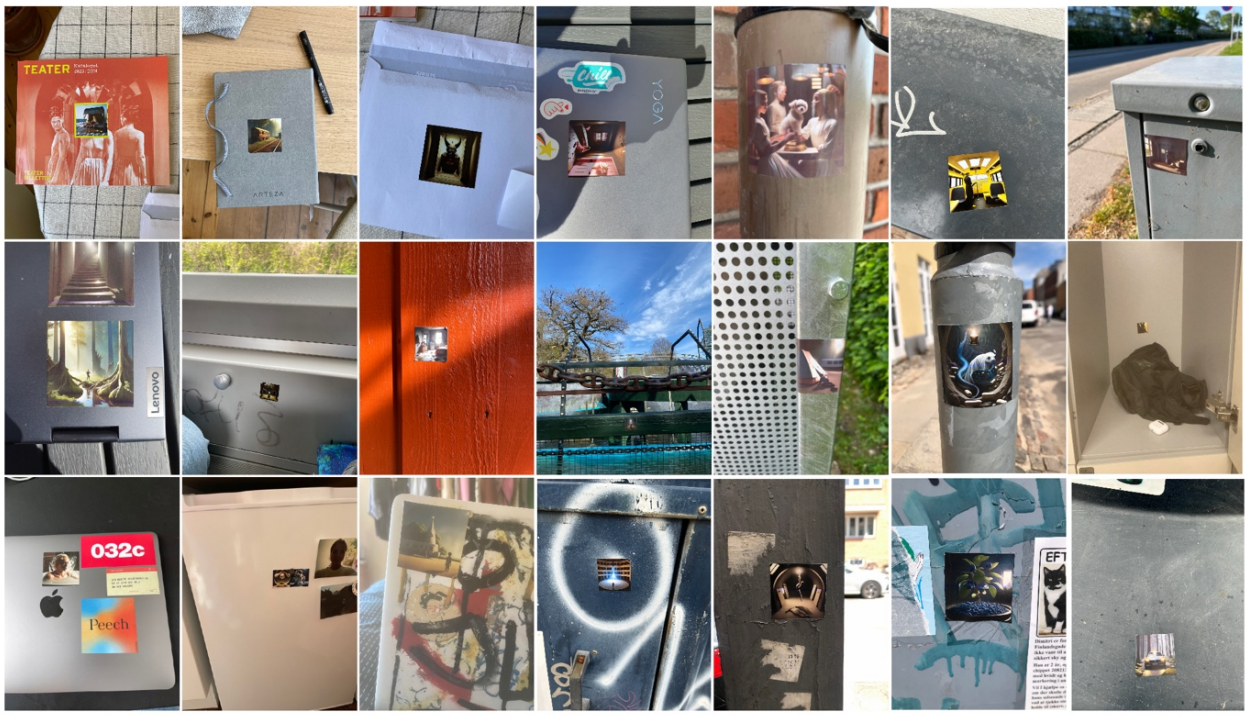


Figure 6: Sticker placements from participants. Pictures sent in by the participants of the probe. Some were sent in through the questionnaire, and some were sent by email. People placed their stickers outside in public, inside their homes, in places they visited, on personal items such as a laptop, and some were saved for later or simply not used at all.

For the participants who placed them on more personal items like a notebook or their laptop, they mentioned how it was meaningful to them because would then carry the sticker with them and remember the dream better. "*I always have my laptop with me, so that's why I chose it. Now there is a meaning behind the stickers I have on my laptop, which there hasn't been for some of the other stickers I have on it"* (P3). Of the people who placed their stickers in public, some people chose places that meant something to them or were a part of their everyday lives to bring their dreams into the world. "*[I placed the stickers in] some of the locations that are important in my life"* (P6). *"I chose to place them in the streets near my home on different items where they probably won't be removed and where I can meet them as I pass by every day. I felt thrilled, maybe because my dream is now a part of this world a little bit more than before, maybe because I let it take place and be important—normally I forget them so quickly"* (P8).

When asked whether they thought about the dream or the sticker after placing it, most participants said that they did. Mostly, people thought about the dream when they saw the sticker they had placed—"*Yes, it made me remember the dreams"* (P4)—but participants mentioned how the image of the sticker became how they remembered the dream, rather than the images they had imagined beforehand: *"Yeah, it came back to me, but more and more it's only this picture and not the whole dream"* (P2), *"Yes! When the sticker pops into my mind or I walk past it and see it, I remember my dream. For me, the sticker has become a sort of poster for my dream, as if it were a movie with a poster or something like that"* (P8).

Most of the participants also spoke to other people about the stickers and enjoyed the conversations they sparked: "[I talked to] my friends because they were there when I put it on my PC" (P1), "Yes, I was with my boyfriend when I put up the stickers, and we talked about the dream and the stickers. It was nice to share!" (P8) Further, some people had also shown the stickers to someone whom they had already told the dream it represented to. "I have spoken to my boyfriend about them, where I have explained what the dreams were about based on the picture. I thought it was fun to show him. Some of the dreams I have tried to tell him about before, so it was fun to show a picture of them" (P3). We had not even considered that it could also be a fun thing to do with others.

Some participants expressed interest in receiving more stickers from each dream, or several options of different AI image generations of the same dream to choose which one represented the dream best: "*maybe I would have been able to feel more connected to the depiction of my dreams if I could choose between more options"* (P2), "*It would probably have been fun to have a few more stickers to choose from for each dream. That way, there is a greater chance that one of them will remind you of the image you also have of your dream in your head"* (P3). Two other participants brought up the fact that the stickers represented key points of the dream, but that there were often changes of scenery and story in a dream: "*It would be nice to also try a version where the stickers describe some of the progress*

in the dream, or if the dream is set in different specific scenes, that the stickers are able to differ between the scenes." (P8). *"Maybe three stickers like in a comic strip so you can better relate to them"* (P6).

Most of the participants were excited about the probe and would do it again—*"Yes! I liked the materialization of my dreams"* (P1)—and noted that this could be an interesting format for remembering dreams more generally: "*Dream stickers would be cool for a dream diary. I like the idea of using them on journals or books"* (P2); "*Yes, I would very much like that. I like to remember my dreams. I sometimes try to write them down, but it would be fun to have a whole collection of dream images. It is almost impossible to remember dreams in the long run, but the pictures would probably help with that"* (P3).

One person specifically enjoyed the aspect of sharing their dreams with the world and said that they would love to do that more. *"Definitely. I love dreaming, but I would love my dreams to also live and take form in this world and not just in my inner world. I would like to share them more. I forget my dreams most of the time, but I can feel that their atmosphere and vibe affect me throughout the day and that it helps when I am able to remember them and understand why my vibe is as it is. The stickers have helped me to remember and just motivated me to write down other dreams, so now I remember them more"* (P8). Only one participant did not want to do it again, because they felt like the stickers did not represent their dreams: "*No, I don't really think the stickers represented my dreams that well" (P6).*

In general, the sticker format seemed to be a productive way to think through what and how dreams might be as a waking design material: "*I chose to place them in the streets near my home, on different items, where they probably won't be removed and where I will meet them when I pass in my every day. I felt thrilled, maybe because my dream now is a part of this world a little bit more than before, maybe because I let it take place and be important - normally I forget them so quickly.*" (P8) Further, placing dreams into the public produced a sense of personal inscription into the public sphere: "*I thought mostly about if there were other stickers. I didn't want to ruin anything, so I selected places where other people had put stickers. My first idea was the electricity thing because I often laugh at some of the stickers there. But I'm curious whether people will notice them and what they will think. It would be fun to observe because they're so specific yet anonymous at the same time. Like if you don't know then there's no context. It's very personal and only me that will know that it's there and what it means. That's kind of special in a way. I look forward to seeing if it's still there in a while. All three spots are on my daily route, so I'll walk by. Normally wouldn't look for stickers outside the electricity thing next to my door but now I will, to see if they're still there"* (P7).

7 DISCUSSION

This project's mixture of methods and approaches—research into contemporary self-tracking categories, large-scale surveys of people's sleep and dream habits, interviewing experts to understand more pointed perspectives on dreams, and a design process—demonstrates a process of trying to articulate the bizarreness of dreams, while not overdetermining dreams themselves. It was important throughout to keep the irrational present to keep from flattening dreaming into rational data. While this study is of a single service of a design fiction concept, the Dream Sticker Machine, and so points to possible future directions for design to be influenced by dreams.

7.1 Dreaming in the everyday

The design probe showed promise in imagining how dreaming can emerge into the world, demonstrating that designing for dreams can lead to new ways to experience them over time—in this case via stickers that inject dreams into the public sphere. Street art practices stem from the wish to reclaim art to exist outside the gallery space [5]. The sticker facilitates new ways of articulating and sharing what takes place during sleep, what kinds of activity we see as productive, and questions who we feel our dreams need to be shared with. While not all of our participants were entirely comfortable using the stickers in public space, as a format that represent dreams in everyday life, they invite a participant to be creative and think out of the box when placing, sharing, or simply saving them. By facilitating a material way of remembering dreams, we sought to initiate introspection, and hopefully create new forms of collective dream-sharing via playful interaction design [35]. While the success of this is not something that is possible to evaluate from this early and short-term deployment, the response from our participants indicates that this kind of avenue for dream representation in public life is a promising direction for future research.

The sticker machine is designed to encourage dreams to take up physical space in people's everyday lives, and a usually ignored or discarded component of human experience becomes visible and discussable in public life. We envision future, longer-term deployment of dream stickers as providing new ways of collaboratively annotating public space—dreams augment the public sphere with the sentiments of those who are in that area while representing their feelings indirectly. As a way of both remembering and sharing dreams, we see the stickers as offering a means of *engaging in collective speculation* on the irrational. An imagined broader social practice of sharing and placing dream stickers promotes a social role for dreaming even as the stickers stand in for an irrational part of our lives we usually forget. We found that externalizing a dream and making it an actual object to carry, place, or pass on forged new ways for remembering dreams for the individual in our probe—as when one probe user said that they placed the stickers places outside they knew they would come across again—but also brings the dream into the public to possibly become a matter of shared consideration. Imagining how this idea could help in other kinds of complex or irreducible data might mean bringing it to be considered by broader groups, where data that eludes straightforward representation or clarity can become the object of many people's thinking. Rather than considering these data as something that is meant for representation and interpretation, we might imagine it as being a form of inspiration and social sensemaking.

7.2 Self-tracking the irrational

In a field where 'self-tracking' broadly demands usefulness, optimization and insight, there is something joyful in subverting those goals. Dreams resist being made useful, even as sleeping is being made to be optimal. Dreams walk the line between being biodata, information created by the body and anecdata, information that

is observed based on personal experience. Walking that line has helped to productively trouble and extend both categories. By embracing the irrationality of dreaming, we produce a frame for self-tracking that builds from Elsden et al.'s idea of the quantified past, even if quantifying dreams itself seems contradictory [30]. Elsden et al offer three aspects for designing for remembering: *experiencing*, *curating,* and *sharing* personal data. While dreams are a kind of data, they are a data that seems to resist these categories—they are pure experience, hard to remember, and nearly impossible to curate or share in themselves. We wanted to avoid reducing such hard-to-define and hard-to-grasp datasets to something overly simplified or too generalizable, as it would lose the connection to the source. The dream sticker machine translates a dream experience into something that can be saved and shared with others on its own terms. Dreams as data resist being reduced to something numerical or tidy—they remain complex, confusing, partial, and incomplete. The stickers offer a way to embrace those aspects of confusing, untidy data and underdetermined situations by embracing them as something that should be celebrated and shared. People are interested in dreams and having them in the world offers the chance to drive new kinds of conversations about them. While dreams are not necessarily true, they are real, and the reality of dreams comes from their relation to the biological activity that produces them, the lived experiences that informs their content, and how a dreamer makes sense—or doesn't—of what they remember of them. The absurd, the strange, and the unusual may offer strong insights into designing for problems and domains where tidier data is normally sought. *Embracing irrational or irreducible components* of problems that are often ignored or bypassed in a design project and transposing them to become a topic of conversation in themselves might help to create new understandings of complex and confusing data and ways of engaging with them.

7.3 AI as a veil between waking and dreaming

The dream sticker machine visualized dreams but did not do it with the goal of representing them accurately to a dreamers' experience. While AI is controversial, the image generator created curiosity about the contents of older dreams in our design probe. Ironically, the differences between the AI-generated images and the dreamer's own memory forces the user to pay more attention to the dreams they might otherwise have forgotten. These images "prompted" our participants' memory of the visual components of their dreams, making the dream more present and memorable in the waking world. By acting as a mediator or interpreter of dreams, the AI imagery helped to create a new form that could be shared in public space. Dreams are immensely personal and supremely private, and the generator creates a translation of the dream that can be spread into the world safely. This project proposes a system that can take your deepest and most personal thoughts and translates them to be acceptable in public. The sticker represents the ideas, concerns, and feelings of a dream, but in a way that does not directly reveal or implicate the dreamer. The dream sticker machine offers one way of shielding privacy in a sensitive area and *using generative veils* might be a way to enliven, obscure, or adapt certain kinds of sensitive data for productive uses in broader contexts.

There are of course huge ethical and practical concerns with using AI technologies more generally—they require ever-larger amounts of energy to train datasets [86], as well as huge volumes of water to cool the datacenters that this training takes place in [70]. Particularly, image generation of the kind used in this project are problematic for both stealing artists' labor wholesale to train models to produce images based on that work [51]. Put differently, generative AI is a disaster ecologically as well as culturally. This project takes the model of generative AI image creation as similar to how dreaming works in humans to drive a design fiction concept. We drew parallels between how AI and dreaming are constructed both metaphorically and practically, in the way large amounts of visual data are scrambled and reconstructed into bizarre and unreal scenarios. However, if we imagine new ways of expressing and representing dream stories, other methods than AI could be a good fit. Sketching in various forms and materials have been productive for many researchers in HCI and design to imagine new forms and futures for different technologies [78]. Watercolor painting, for example, has been productively used to react to ethnographic findings [56], to layer and annotate images in conversation [32], as well as to speculate towards new technologies where the "leaky boundaries" of watercolor as a medium helps to create emergent ambiguities and open new possibilities in representation [87]. While these avenues offer something different than a disintermediated and seamless AI representation of a dream and might weaken—or at least change—the idea of a generative veil, working with dreams across various mediums is an exciting direction for future research.

8 CONCLUSION

Dreams are contradictory, individual, and hard to grasp. They are also universal, inspiring, and a frequent source of fascination for people. We began this project looking at current sleep technologies and questioning the usual approach of sleep optimisation that focuses on achieving 'a good night's sleep'. Through a questionnaire, we found that people's relation to their dreams are highly individual. For some, dreams have no specific meaning, while others see them as a reflection of their mental state and innermost thoughts and feelings. Dreams can be private, something we want to share, or something we use productively for solving problems in our everyday lives or for enabling creative thinking.

As part of a research through design project, we worked with dreams as a design material to develop new ways of thinking of and talking about dreams in HCI. The 'dream sticker machine' explores visualising dreams through an AI image generator, creating a sticker as a form of interaction and communication to aid in making dreams something to share and reflect on in a playful and creative way. The sticker machine proposes a way of using interactive technology to support potential users in remembering their dreams better, but also facilitates repeating moments of introspection and new ways of sharing dreams in public spaces. Working with dreams as material in the context of interaction design research has been a vantage point for appreciating abstract and somewhat anecdotal data derived from our sleeping bodies adapted into the waking world. We see potential in using dreaming to understand designing with other forms of underdetermined, and confounding data by engaging in collective speculation, embracing and expressing the

irrational, and creating veils for data to make expressing deeply held feelings safer.

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